Chp ~ 6

Electromagnetic Induction.

⟹ Magnetic Flux. (weber).

If we consider a plane perpendicular to a uniform magnetic field then the product of magnitude of the field and the area of the plane is called magnetic flux (ϕ).

$$\phi = \underbrace{B}_{\text{magnetic field}} \; \underbrace{A}_{\text{Area}} \cos \underbrace{\theta}_{\text{Angle between Area vector and B}}$$

$$\phi = \vec{B} \cdot \vec{A}$$

$$\phi = BA \cos 90^\circ = 0 \quad \text{(minimum).}$$

$$\phi = BA \cos 0 = BA \quad \text{(maximum).}$$

$$\underbrace{B}_{\text{weber}/m^2} = \frac{\phi}{A}$$

$$\phi = \frac{F}{IL} \times A.$$

$$\boxed{\phi = [ML^2T^{-2}A^{-1}]}$$

⟹ Faraday's Law of Electromagnetic Induction.

i] First Law: when the magnetic flux through a circuit is changing an induced emf is set up in the circuit.

Whose magnitude is equal to negative rate of change of magnetic flux.

(Neumann's Law).

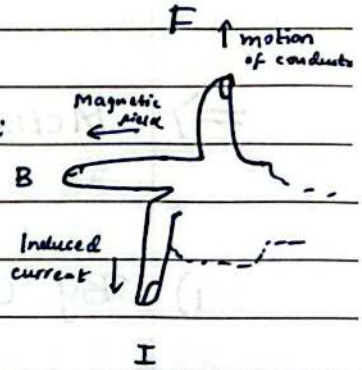
$$\boxed{\mathcal{E} = -\frac{d\phi}{dt}} \quad \text{or} \quad \boxed{\mathcal{E} = -\frac{d}{dt}\Sigma\phi}$$

$$\phi = \vec{B} \cdot d\vec{A}$$

ii] Second Law: The direction of induced emf or the current in any circuit is such as to oppose the cause that produces it. (Also known as Lenz Law)

To find direction of induced current:

Flemings Right Hand Rule.

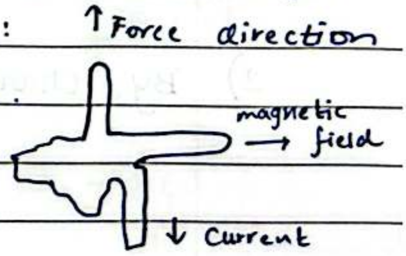


(FBI open up !!!)

(FBI).

To find direction of magnetic force:

Flemings Left Hand Rule.



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⟹ Induced Current and Induced Charge.

$$\mathcal{E} = -\frac{d\phi}{dt}$$

$$\mathcal{E} = -N\frac{d\phi}{dt}$$

Now, $I = \frac{\mathcal{E}}{R}$

$$I = -\frac{N}{R}\frac{d\phi}{dt}$$

Now, $I \times dt = q.$

$$q = \frac{-N}{R}\frac{d\phi}{\cancel{dt}} \times \cancel{dt}$$

$$\boxed{q = \frac{-N}{R} \times d\phi}$$

⟹ Methods of Changing Magnetic Flux.

$(\phi_B = BA\cos\theta)$

1) By Changing B : $\mathcal{E} = -\frac{d}{dt}(BA\cos\theta)$.

$$\boxed{= A\cos\theta \times -\frac{dB}{dt}.}$$

2) By Changing A : $\mathcal{E} = -\frac{d}{dt}(BA\cos\theta)$

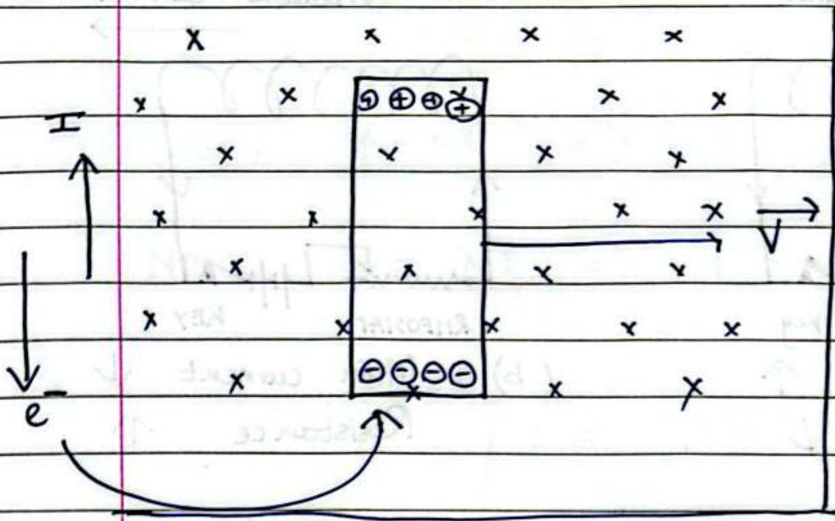
$$\boxed{\mathcal{E} = -B\cos\theta \times \frac{dA}{dt}.}$$

3) By changing θ : $\mathcal{E} = -\frac{d}{dt}(BA\cos\theta)$

$= -BA\frac{d}{dt}(\cos\theta) \Rightarrow -BA \times (-\sin\theta) \times \frac{d\theta}{dt}$

$$\boxed{\mathcal{E} = -BA\sin\theta \times \omega}$$ Angular velocity

⇒ Motional EMF (Motion of a conductor in magnetic field).



$F_e = F_m$

$\cancel{q}E = \cancel{q}vB$

$E = vB.$

$(E = V/l)$

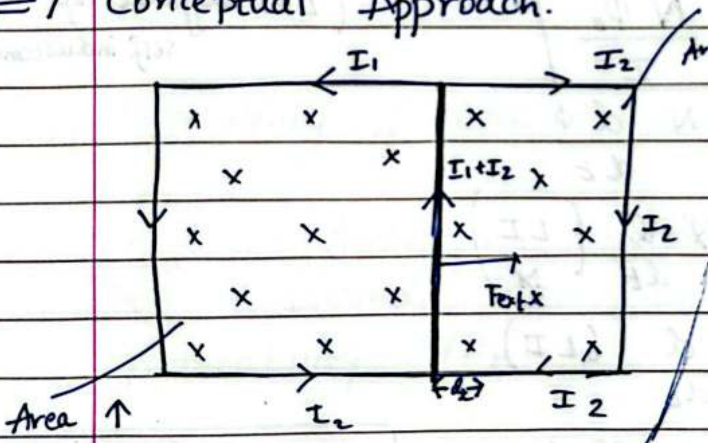
$\frac{V}{l} = vB$

$\boxed{V = Bvl}$

magnetic field, velocity, length.

When a straight conductor of length l moves with a velocity 'v' perpendicular to a magnetic field 'B' an emf (or P.D) is developed across the ends of the conductor which is known as Motional emf.

⇒ Conceptual Approach.



$F = -IBl.$

$dW = f \times dx$

$\varepsilon \times \cancel{I}dt = -\cancel{I}Bl\, dx$

$\varepsilon = -lB\left(\frac{dx}{dt}\right)$

$\boxed{\varepsilon = -lBv}$

Now, ⇒ $I = \frac{\varepsilon}{R}$

$I = \frac{-Blv}{R}.$

$V = W/q$

$W = Vq$

$dW = \varepsilon \times I dt$

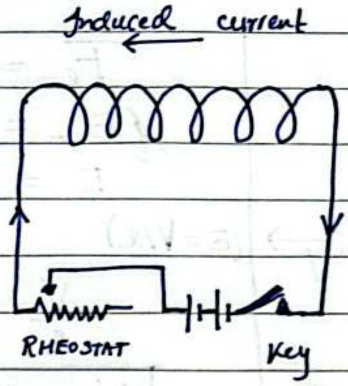
Now $P = f \times v. = IBl \times v.$

$= \frac{Blv}{R} \times Bl \times v$

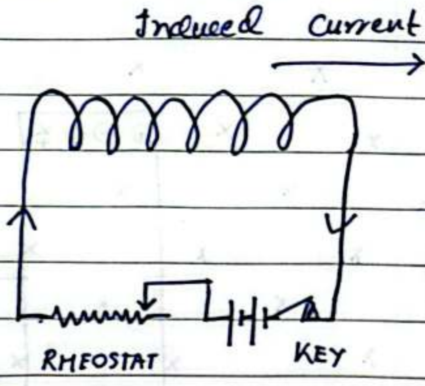
$= \frac{B^2 l^2 v^2}{R}$

$\Rightarrow \boxed{P = \frac{B^2 l^2 v^2}{R}}$

⇒ Self Induction 'L'.



(a) Main Current ↑
Resistance ↓



(b) Main current ↓
Resistance ↑

The phenomenon of electromagnetic Induction in which on changing the current in coil an opposing induced emf is set up in that very coil is called self Induction.

$$N\phi_B \propto I$$

$$N\phi_B = LI$$

$$\boxed{L = \frac{N\phi_B}{I}}$$

($L \Rightarrow$ coefficient of self inductance)

Now,
$$\mathcal{E} = -N\frac{d\phi}{dt}$$

$$\mathcal{E} = -\cancel{N}\frac{d}{dt}\left(\frac{LI}{\cancel{N}}\right)$$

$$\mathcal{E} = -\frac{d}{dt}(LI)$$

$$\boxed{\mathcal{E} = -L\frac{dI}{dt}} \quad \text{or} \quad \boxed{L = \frac{-\mathcal{E}}{dI/dt}}$$

Self Inductance : is equal to no of flux linkages when current is passing through coil A.

⟹ Self Inductance of a long Solenoid.

$$B = \mu_0 \frac{N}{l} I.$$

Now, $\phi = BA \Rightarrow \phi = \mu_0 \frac{N}{l} I \times A.$

Now, $L = \frac{N\phi}{I} \Rightarrow L = \frac{N}{\cancel{I}} \times \frac{\mu_0 N \cancel{I} A}{l \times \cancel{I}}$

$$\boxed{L = \frac{\mu_0 N^2 A}{l}.}$$

If there is a medium 'K'

$$\boxed{L = \frac{\mu_0 N^2 A K}{l}}$$

⟹ Factors on which self Inductance depends.

i] Number of turns : $L \propto N^2$. (directly proportion)

ii] Area of cross section : $L \propto A$. (directly proportional)

iii] Permeability of core material : $L \propto \mu_r$

Self Inductance becomes μ_r times, If it is wound over an core who has permeability (μ_r).

⟹ SI unit is Henry

⟹ Energy stored in an Inductor.

Energy supplied by battery is stored in Inductor in form of P. E

$$\mathcal{E} = -L\frac{dI}{dt}$$

Now, $dw = -\mathcal{E} \times dq$. $(w = V \times q)$

$$dw = -L\frac{dI}{dt} \times dq$$

$$\int dw = -L\int I\,dI$$

$$w = -L\frac{I^2}{2}$$

$$\boxed{U = \frac{1}{2}LI^2}$$

$$\frac{U}{V} = \frac{1}{2}\frac{B^2}{\mu_0}$$ ⟹ Energy Density (Energy stored per unit Volume)

⟹ Mutual Induction

$N_2\phi_2 \propto I_1$

$N_2\phi_2 = MI_1$

$$\boxed{\frac{N_2\phi_2}{I_1} = M}$$

$$\mathcal{E}_2 = -N_2\frac{d\phi_2}{dt}$$

$$\mathcal{E}_2 = -\cancel{N_2}\frac{d\phi I_1}{dt\,\cancel{N_2}}$$

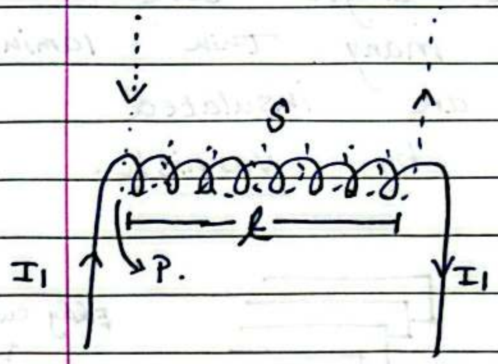
$$\mathcal{E}_2 = -M\frac{dI_1}{dt}$$

$$\boxed{\frac{\mathcal{E}_2}{-dI_1/dt} = M}$$

⟹ When we place two coils near each other and pass electric current in one of them then an emf is produced in the second coil.

This is called mutual induction

⇒ Mutual Induction of two long coaxial Solenoids.



$$B = \mu_0 \frac{N_1}{l} \times I_1$$

$$\phi_2 = BA = \mu_0 \frac{N_1}{l} \times I_1 \times A.$$

Now, $$M = \frac{N_2 \phi_2}{I_1}$$

$$M = \frac{N_2}{\cancel{I_1}} \times \frac{\mu_0 N_1 \cancel{I_1} A}{l}$$

$$\boxed{M = \mu_0 \frac{N_1 N_2 A}{l}}$$

$$\left(n_1 = \frac{N_1}{l} \right)$$

$$\boxed{M = \mu_0 n_1 N_2 A}$$

⇒ Eddy Currents / Foucault Currents.

It is observed that when a piece of metal is placed in a changing magnetic field causing a change in magnetic flux linked with it, the induced currents are set up throughout the volume of metal which opposes the motion of metal.

→ This type of Induce current is Eddy currents. It depends upon resistance of metal

If $R >>>$ Eddy currents are feeble.

→ In Transformers they are undesirable as

i] large heat produce (loss of energy).

ii] Flux produced by them.

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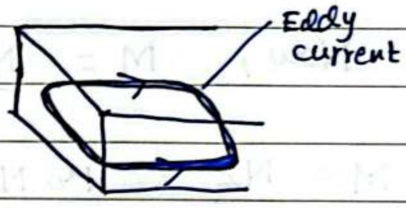
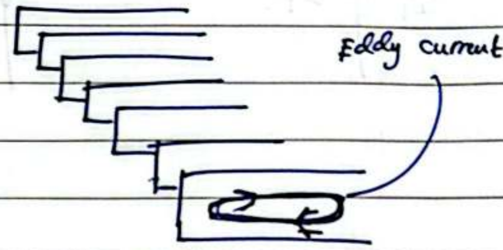
→ To minimise these currents the currents are not taken as a single piece core but are made of many thin laminas of soft-iron which are insulated from each other by <u>varnish</u>.



a) Solid core

R —— ↓

I. —— ↑.



b) Laminated core.

R ⟶ ↑

I ⟶ ↓